Advanced Types of Machine Learning

Semi-Supervised and Reinforcement Learning Explained

Exploring sophisticated approaches that go beyond traditional supervised learning methods

Semi-Supervised Learning: The Best of Both Worlds

What It Is

Semi-supervised learning is a hybrid approach that combines the strengths of both supervised and unsupervised learning. It leverages a **small amount of labelled data** alongside a **large volume of unlabelled data** to build more robust models.

The Power of Mixing

This technique is particularly valuable when labelling data is expensive, time-consuming, or requires domain expertise. The unlabelled data helps the model learn underlying patterns and structures, whilst the labelled examples guide it towards accurate predictions.

Example in Action: Consider 1,000 animal images where only 100 are labelled as "cat" or "dog", whilst 900 remain unlabelled. The model learns from both sets to accurately classify future images.

Semi-Supervised Learning Techniques

Self-Training

Model iteratively labels its own predictions with high confidence, gradually expanding the training set

Label Propagation

Spreads labels from labelled to unlabelled data points based on similarity measures

Graph-Based Methods

Represents data as graphs where nodes are samples and edges indicate similarity

Autoencoders & GANs

Deep learning architectures that learn compressed representations from unlabelled data

Why This Matters

- Data labelling is expensive and time-intensive
- Unlabelled data is abundant and accessible
- Performance improvements without proportional cost increases

Semi-Supervised Learning in Practice

Medical Image Classification

Analysing X-rays, MRI scans, and CT images where expert annotations are limited but unlabelled scans are plentiful

Speech Recognition

Training models on vast audio datasets with minimal transcriptions, improving accuracy across diverse accents and languages

Fraud Detection

Identifying suspicious transactions using few confirmed fraud cases alongside millions of regular transactions

Web Content Categorisation

Organising and classifying vast amounts of online content with limited human-curated examples

Key Benefit: Semi-supervised learning achieves significantly improved accuracy whilst dramatically reducing labelling costs and time investment.

Reinforcement Learning: Learning Through Experience

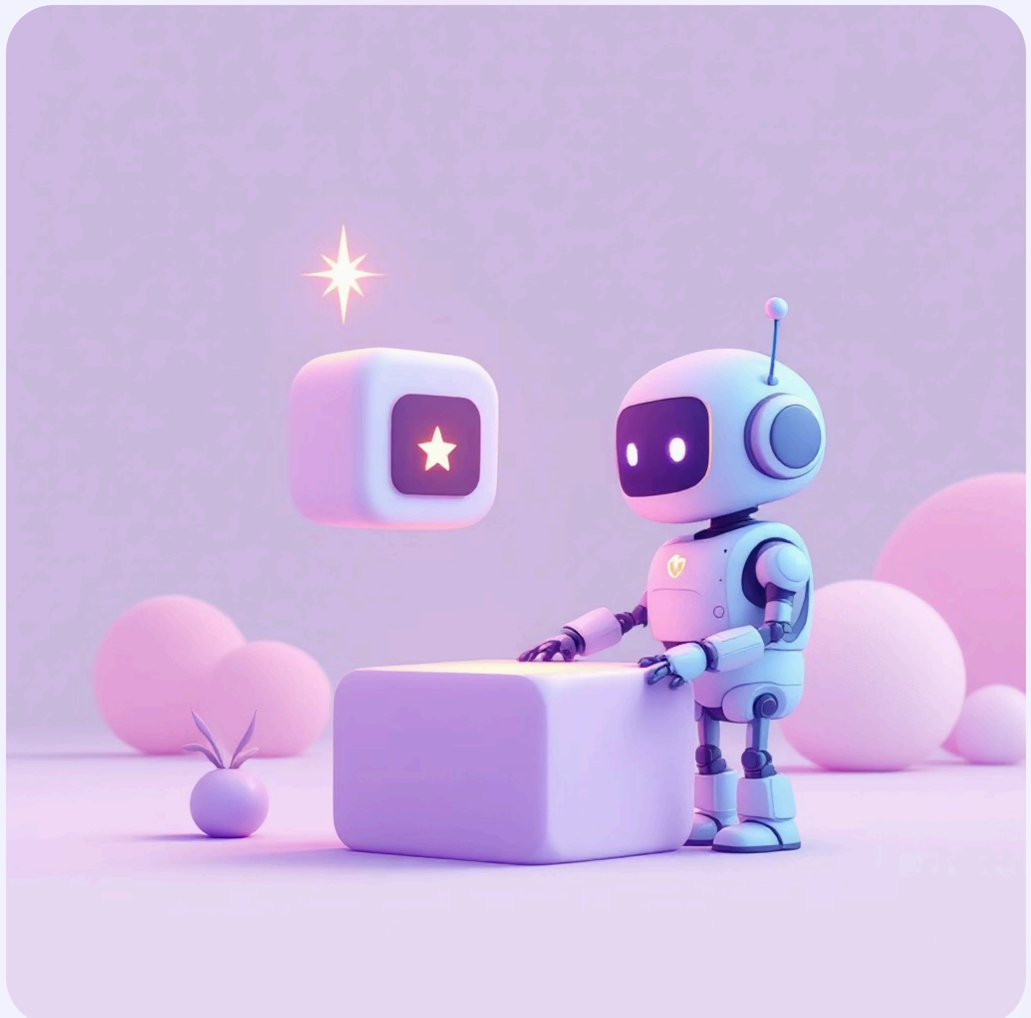
The Core Concept

Reinforcement Learning (RL) is fundamentally different from other ML paradigms. Instead of learning from pre-existing datasets, an RL agent learns through interaction and feedback with its environment.

The model takes actions, receives rewards or penalties, and adjusts its strategy to maximise cumulative rewards over time.

The Learning Loop

The agent continuously cycles through observation, action, and feedback—gradually discovering optimal strategies through trial and error.



1. **Agent Takes Action**

   Decides what to do based on current state

2. **Environment Responds**

   Provides feedback through rewards or penalties

3. **Agent Learns & Adapts**

   Updates strategy to maximise future rewards

Classic Example: A robot learning to walk—each successful step earns +1 reward ✅, whilst falling results in −1 penalty ❌. Over time, it discovers optimal movement patterns.

Core Components of Reinforcement Learning

Component	Description
Agent	The learner or decision-maker that interacts with the environment and takes actions to achieve goals
Environment	The external system or world that the agent operates within, providing states and responding to actions
Action	The set of all possible moves or decisions the agent can make in any given state
Reward	The immediate feedback signal (positive or negative) that evaluates the quality of an action
Policy	The agent's strategy or rule that maps states to actions, determining how it behaves

These components work together in a continuous cycle, enabling the agent to learn optimal behaviours through experience. The **policy** evolves over time as the agent accumulates knowledge about which actions yield the highest **rewards** in different situations.

Popular Reinforcement Learning Algorithms

1

Q-Learning

A value-based, model-free algorithm that learns the quality (Q-value) of state-action pairs. It uses a Q-table to store values and updates them using the Bellman equation.

- Off-policy learning approach
- Works well for discrete action spaces
- Foundation for more advanced methods

2

Deep Q-Network (DQN)

Extends Q-learning by using deep neural networks to approximate Q-values, enabling handling of high-dimensional state spaces like images.

- Experience replay for stable training
- Target network for convergence
- Breakthrough in game-playing AI

3

Policy Gradient

Directly optimises the policy by computing gradients that increase the probability of actions leading to higher rewards.

- Handles continuous action spaces
- More stable in certain environments
- Includes variants like REINFORCE and PPO

4

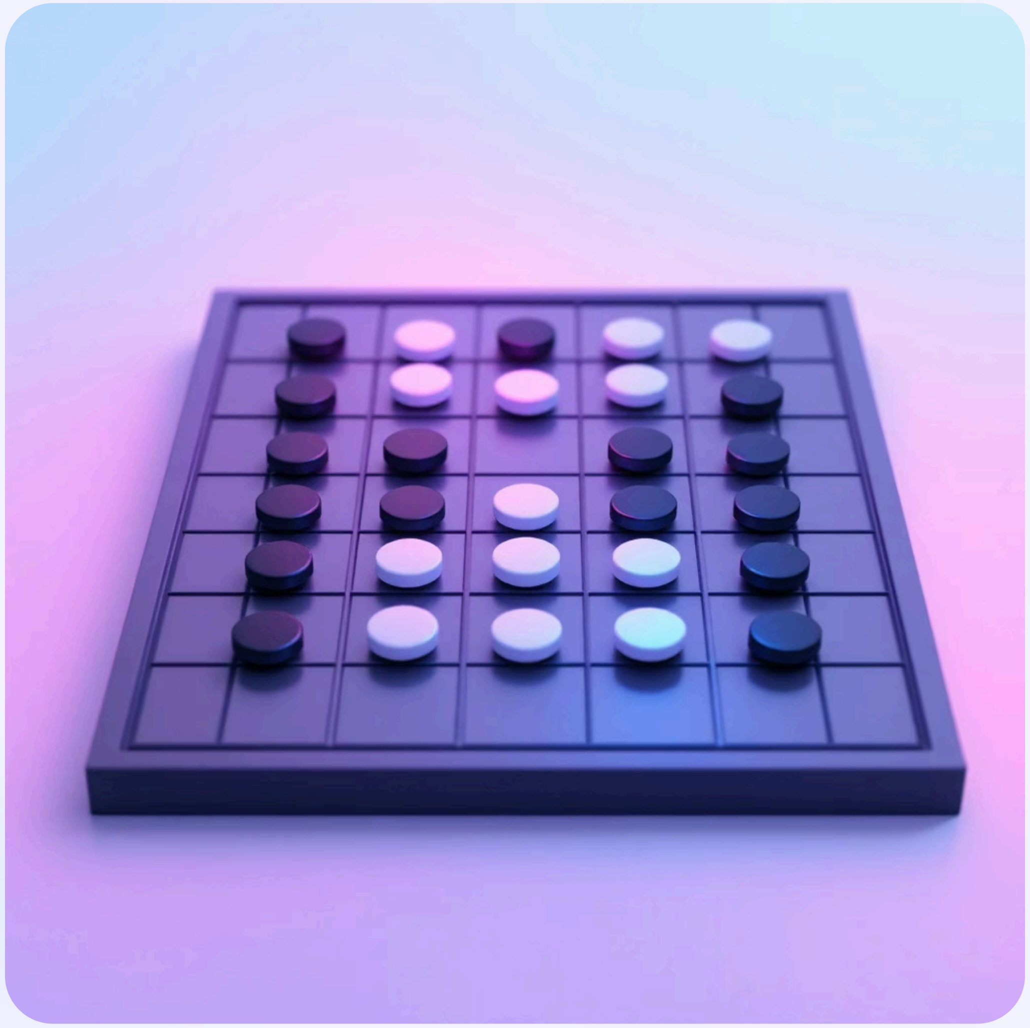
SARSA

State-Action-Reward-State-Action algorithm that learns on-policy, updating based on the action actually taken rather than the optimal action.

- On-policy learning method
- More conservative than Q-learning
- Better for safety-critical applications

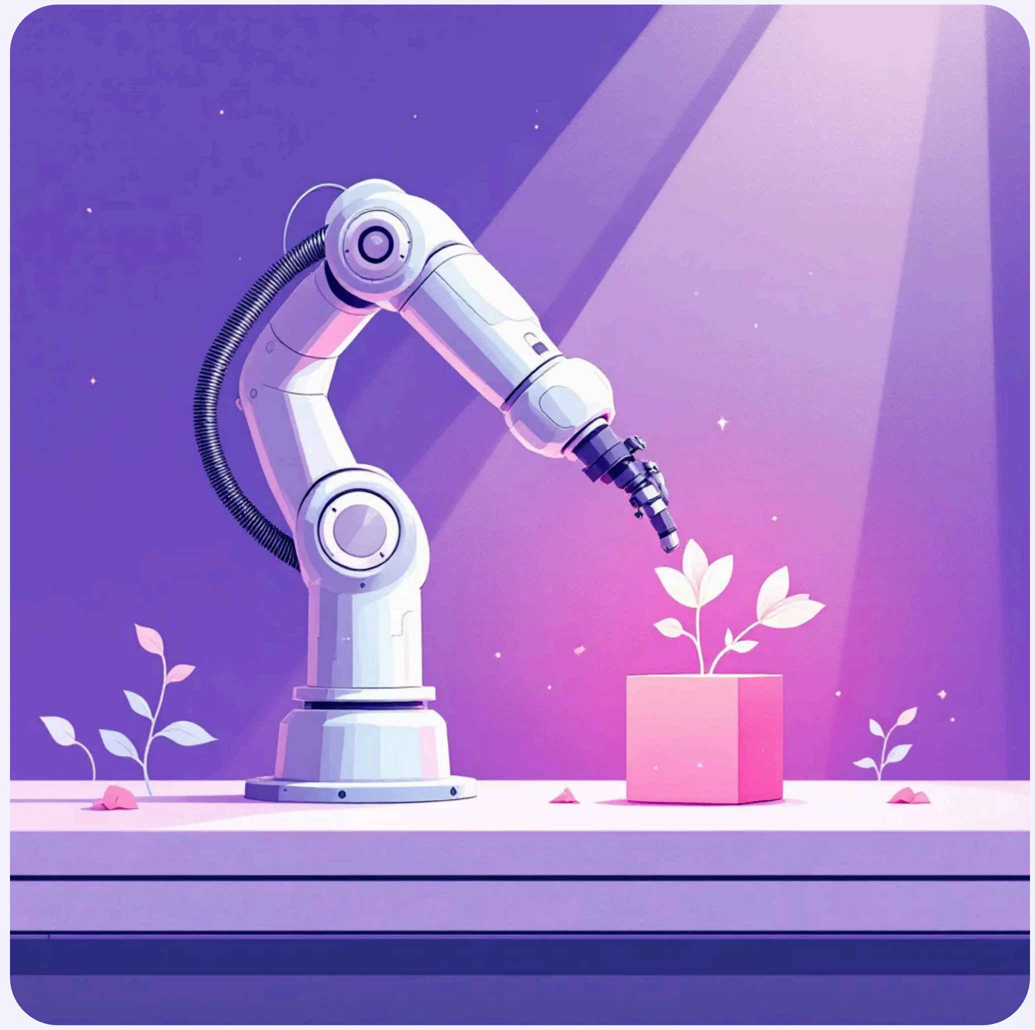
> The ultimate goal of all RL algorithms: Maximise cumulative reward through continuous experience and intelligent exploration

Real-World Applications of Reinforcement Learning



Game AI & Strategic Play

From mastering chess to defeating world champions in Go (AlphaGo) and achieving superhuman performance in Atari games—RL has revolutionised game-playing AI.



Robotics & Movement Control

Teaching robots to navigate complex environments, manipulate objects with precision, and perform intricate tasks through trial-and-error learning.

Autonomous Vehicles

Self-driving cars use RL to make split-second decisions about acceleration, braking, and steering whilst navigating dynamic traffic conditions and optimising for safety.

Dynamic Pricing

E-commerce platforms and ride-sharing services employ RL to adjust prices in real-time based on demand, competition, and market conditions to maximise revenue.

Recommendation Systems

Streaming services and online retailers use RL to personalise content suggestions, learning from user interactions to improve engagement and satisfaction over time.

Comparing Machine Learning Paradigms

Type	Labelled Data?	Primary Goal	Key Algorithms	Common Applications
Supervised	✅ Yes	Predict outputs from inputs	Linear Regression, Random Forest, SVM	Spam detection, price prediction, image classification
Unsupervised	❌ No	Discover hidden structure	K-Means, PCA, Hierarchical Clustering	Customer segmentation, anomaly detection, dimensionality reduction
Semi-Supervised	⚫ Partially	Leverage few labels effectively	Self-training, Label Propagation, Autoencoders	Medical imaging, speech recognition, web content categorisation
Reinforcement	❌ No (uses rewards)	Learn optimal behaviour through experience	Q-Learning, DQN, Policy Gradient, SARSA	Game AI, robotics, autonomous vehicles, dynamic pricing

Each paradigm addresses different learning scenarios and data availability constraints. The choice depends on your problem structure, data resources, and desired outcomes.

Key Takeaways: The ML Learning Spectrum

Different Paradigms, Different Approaches

ML types fundamentally differ in **how models learn from data**—from direct supervision to autonomous exploration

Supervised Learning

Learns from explicitly labelled examples, ideal when clear input-output pairs are available

Unsupervised Learning

Explores unlabelled data to uncover hidden patterns and structures without guidance

Semi-Supervised Learning

Strategically combines both approaches, maximising value from limited labels

Reinforcement Learning

Learns through trial, error, and feedback, discovering optimal strategies independently

> *"Machines learn best—not when told what to do, but when allowed to learn what works."*

Understanding these paradigms empowers you to select the right approach for your specific challenges, unlocking the full potential of machine learning in diverse domains.